

Dyadic regression

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Abstract

Dyadic regression forecasts a series from its last value, the sum of the two values before that, the sum of the four before those, and so on, with one coefficient per block. With L lags it fits $\log_2(L + 1)$ coefficients. It represents every trailing moving average of length $2^m - 1$ exactly, and it approximates any power-law lag profile $k^{-\alpha}$ to within relative error $2^{\alpha/2} - 1$ uniformly in L . Weighting each lag by $1/(2k - 1)$ inside its block cuts that error to about $2^{|\alpha-1|/2} - 1$, from 0.62 to 0.15 for fractional differencing at $d = 0.4$, and makes the all-equal point exactly the harmonic forecaster of Fishelson and Mohri (2026).

1 The model

Partition the lags $k = 1, \dots, L$, with $L = 2^J - 1$, into dyadic blocks

$$G_j = \{2^j, \dots, 2^{j+1} - 1\}, \quad j = 0, \dots, J - 1,$$

so $G_0 = \{1\}$, $G_1 = \{2, 3\}$, $G_2 = \{4, \dots, 7\}$. Dyadic regression is the forecast

$$\hat{y}_{t+1} = \sum_{j=0}^{J-1} \theta_j S_j(t), \quad S_j(t) = \sum_{k \in G_j} y_{t+1-k}, \quad (1)$$

an autoregression whose coefficient on lag k is $\theta_{j(k)}$, shared within each block. Writing the regressors as block averages $\bar{S}_j = S_j/2^j$ instead gives the same model with coefficients $2^j \theta_j$. The `grouped_ar` transform of the `skaters` library implements (1) with block sums, fitted by recursive least squares with exponential forgetting and a ridge toward zero.

The number of coefficients is $J = \log_2(L + 1)$, against L for an unrestricted autoregression. The question is what the restriction costs and what it buys.

2 What the blocks can represent

Moving averages. The trailing average of length $W = 2^m - 1$ is exactly $\theta_j = 1/W$ for $j < m$ and $\theta_j = 0$ for $j \geq m$. Every other window length differs from one of these by part of a single block. Dyadic regression therefore contains, exactly or nearly, the moving average at every time scale, with one coefficient per doubling. This is the same covering that lets online learners compete with the best window of any length using $O(\log T)$ experts (Daniely et al., 2015).

Power-law profiles. Long memory shows up as autoregressive coefficients that decay like a power of the lag. For fractional differencing $(1 - B)^d y_t = \varepsilon_t$ with $0 < d < \frac{1}{2}$ the coefficients of the $\text{AR}(\infty)$ representation satisfy $\pi_k \sim C k^{-(1+d)}$.

Proposition 1. *Let $\varphi_k = c k^{-\alpha}$ with $\alpha > 0$. Choosing θ_j as the geometric mean of φ at the two ends of block G_j gives, for every L and every lag $k \leq L$,*

$$\left| \frac{\theta_{j(k)}}{\varphi_k} - 1 \right| \leq 2^{\alpha/2} - 1.$$

Consequently, for any series, the one-step forecast of the dyadic model differs from that of the full $\text{AR}(L)$ model by at most $(2^{\alpha/2} - 1) \sum_{k \leq L} |\varphi_k| |y_{t+1-k}|$.

Proof. On G_j the ratio of the largest to the smallest value of φ is $((2^{j+1} - 1)/2^j)^\alpha < 2^\alpha$. The geometric mean θ_j of the two extremes is within a factor $2^{\alpha/2}$ of every value in the block, so $\theta_j/\varphi_k \in [2^{-\alpha/2}, 2^{\alpha/2}]$, and $1 - 2^{-\alpha/2} \leq 2^{\alpha/2} - 1$. The forecast bound follows from the triangle inequality. $\square$

The bound does not shrink as L grows, and it is not small: 0.23 at $\alpha = 0.6$, 0.41 at $\alpha = 1$, and 0.62 for fractional differencing at $d = 0.4$. It is a worst case over lags, and the least-squares coefficients of a fitted model trade it against the variance saving of $L - J$ fewer parameters. Blocks growing by a ratio $r < 2$ instead of 2 cut the error to $r^{\alpha/2} - 1$ at the cost of $\log_r(L)$ coefficients. Table 1 gives the numbers.

profile	$r = 2$ (13 blocks)	$r = 1.5$ (21)	$r = 1.25$ (38)
$k^{-0.6}$	0.231	0.129	0.069
k^{-1}	0.414	0.225	0.118
$k^{-1.4}$	0.624	0.328	0.169
$(1 - B)^{0.2}, \pi_k$	0.516		
$(1 - B)^{0.4}, \pi_k$	0.624		

Table 1: Worst relative error of the best one-coefficient-per-block approximation over lags $k \leq 4096$, for geometric blocks of ratio r . Dyadic regression is $r = 2$. The fractional-differencing rows use the exact $\text{AR}(\infty)$ coefficients and match the bound of Proposition 1 with $\alpha = 1 + d$. Computed by `check_dyadic.py`.

This is the dyadic form of the argument for the heterogeneous autoregressive model of Corsi (2009), which regresses on daily, weekly and monthly averages and reproduces long memory with three coefficients. Hwang and Shin (2014) show that the infinite-order version is a long-memory model exactly when its coefficients decrease exponentially across horizons, and that a fitted finite-order version loses little to truncation. Dyadic regression fixes the horizons at the powers of two and lets the data choose the coefficients.

3 Harmonic dyadic regression

Most of the error in Proposition 1 comes from treating every lag in a block alike when the profile falls across the block. Weighting lag k by $1/(2k - 1)$ inside its block,

$$\hat{y}_{t+1} = \sum_{j=0}^{J-1} \theta_j H_j(t), \quad H_j(t) = \sum_{k \in G_j} \frac{y_{t+1-k}}{2k - 1}, \quad (2)$$

keeps one coefficient per block and puts the profile’s typical decay into the regressor. A power law $c k^{-\alpha}$ then has to be matched only up to the residual factor $k^{1-\alpha}$, which varies by at most $2^{|\alpha-1|}$ across a block. The argument of Proposition 1 gives a worst relative error of $2^{|\alpha-1|/2} - 1$ for weights $1/k$. The weights $1/(2k - 1)$ add a small correction at short lags, 0.054 at $\alpha = 1$ where the bound is zero. For long memory, $\alpha = 1 + d$, the error is $2^{d/2} - 1$: 0.035 at $d = 0.1$ and 0.149 at $d = 0.4$, against 0.46 and 0.62 for plain dyadic regression. Table 2 gives the numbers.

profile	dyadic	harmonic dyadic
$k^{-0.6}$	0.231	0.152
k^{-1}	0.414	0.054
$k^{-1.4}$	0.624	0.149
k^{-2}	1.000	0.414
$(1 - B)^{0.1}$	0.464	0.035
$(1 - B)^{0.2}$	0.516	0.072
$(1 - B)^{0.3}$	0.569	0.109
$(1 - B)^{0.4}$	0.624	0.149

Table 2: Worst relative error over lags $k \leq 4096$, thirteen blocks in both columns. Computed by `check_dyadic.py`.

With every θ_j equal, (2) is a weighted average of past values with weight $1/(2k - 1)$ on lag k . That is exactly the stored forecast of Fishelson and Mohri (2026), discussed next.

4 The equal-weight point

Set every coefficient on the block averages equal, which is $\theta_j \propto 2^{-j}$ on the sums. Lag k in block j then gets weight $w_k = 2^{-j}$, and since $2^j \leq k < 2^{j+1}$,

$$\frac{1}{k} \leq w_k < \frac{2}{k}.$$

The profile is $1/k$ up to a factor of two, a staircase with one step per doubling. Every dyadic scale gets the same total weight, and the weights over L lags sum to $\log_2(L + 1)$.

Fishelson and Mohri (2026) forecast with exactly this shape. Their algorithm stores a weighted average of past outcomes with weights $1/(2k - 1)$ by age, and announces a mixture of its past stored averages with the same weights. They prove that it is calibrated against every outcome sequence, with average calibration error falling like $1/\log T$, and that the kernel $1/(2(t - s) - 1)$, the discrete Hilbert transform, is the best choice up to a constant. The reason is that a forecast’s calibration error is a comparison between the outcomes that built it and the outcomes on which it is used, and a kernel with equal weight at every scale offers an adversary no scale to attack.

For harmonic dyadic regression the match is exact. With all θ_j equal and normalized, (2) is their stored average p_t . Their calibrated forecast is one step further, the same harmonic mixture of the stored averages p_s , $s < t$, and the guarantee is for that mixture. For plain dyadic regression the equal-weight point is the staircase above.

Conjecture 1. *The Fishelson and Mohri (2026) guarantee holds, with a larger constant, when their harmonic kernel is replaced by the dyadic staircase.*

Their own remark that the weights $1/k$ give the same rates as $1/(2k - 1)$ suggests the result does not depend on the exact kernel, and dyadic analogues of the Hilbert transform are classically bounded. Their proof, however, uses moment bounds specific to the Hilbert matrix, so the staircase needs its own check. Harmonic dyadic regression avoids the question.

Remark 1 (Fitted versus frozen). *Fitting the coefficients is what makes dyadic regression forecast well on real series, where some scales predict and others do not. Freezing them at the equal-weight point is what makes the Fishelson and Mohri (2026) forecaster robust, since a learned preferred scale is exactly what an adversary exploits. A frozen forecaster that uses only past outcomes is calibrated but carries no information about the input. The two can be combined by shrinking the fitted coefficients toward the equal-weight point instead of toward zero, so the model starts from the forecaster no sequence can exploit and moves only as far as the data justify. Whether the shrunk model keeps any form of the guarantee is open.*

5 Summary

Dyadic regression is an autoregression with one coefficient per doubling of the lag. It contains the moving average at every scale and approximates power-law memory to a fixed relative error with $\log_2 L$ coefficients. Weighting lags by $1/(2k - 1)$ inside each block keeps the same number of coefficients, cuts the long-memory error by a factor of four or more, and makes the all-equal point exactly the scale-free forecaster that online calibration theory singles out. The numbers in this note are reproduced by `check_dyadic.py`.

References

- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics* 7(2), 174–196.
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